Data Science Capstone Project Report

Predicting SpaceX Falcon 9 First-Stage Landing Success

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GitHub: https://github.com/Downey112/ibm-ds-capstone-spacex

Outline

- Executive Summary
- Introduction
- Methodology
- Results
  - Exploratory data analysis (visualisation and SQL)
  - Interactive analytics (Folium map, Plotly Dash)
  - Predictive analysis (classification)
- Conclusion
- Appendix

Executive Summary

- Methods: SpaceX REST API + Wikipedia web scraping → data wrangling → EDA with charts and SQL → Folium map + Plotly Dash → 4 classifiers tuned with GridSearchCV
- Data: 90 Falcon 9 launches (Falcon 1 removed); overall landing success 66.7% (60/90)
- Trend: landing success rose from 0% (2010–2013) to 90% (2019) and 84% (2020)
- Drivers: flight experience, orbit type, launch site and booster version
- Models: Logistic Regression, SVM, Decision Tree and KNN all reach 83.3% test accuracy (15/18); Decision Tree has the best cross-validation score (86.3%)
- Main error type: false positives (predicting a landing that failed)

Introduction

- Background: SpaceX advertises Falcon 9 launches at ~$62 million; other providers charge upward of $165 million. Much of the saving comes from reusing the first stage.
- Why it matters: if we can predict whether the first stage will land, we can estimate the cost of a launch. That helps a competitor bidding against SpaceX.
- Problem: can mission features (payload mass, orbit, launch site, flight number, booster details) predict a successful first-stage landing?
- Questions:
  - Which factors drive landing success?
  - Which sites and orbits have the best record?
  - Which classification model predicts best?

Methodology

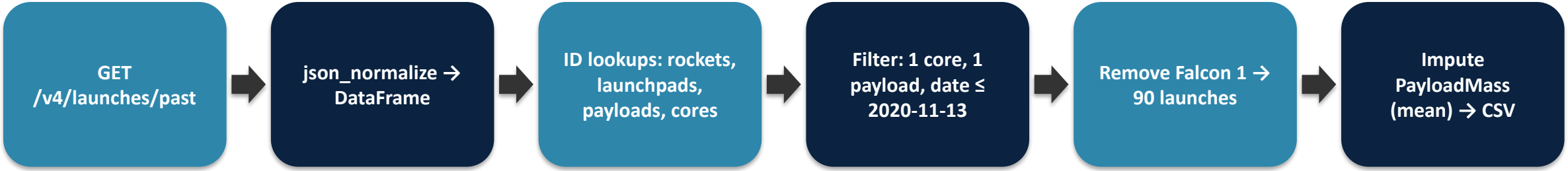
Data collection · wrangling · EDA · interactive analytics · predictive analysis

Methodology Overview



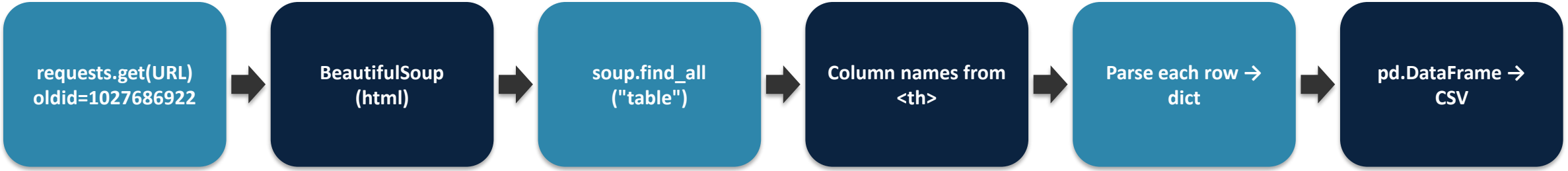
- Data collection: SpaceX REST API v4 and BeautifulSoup scraping of Wikipedia
- Data wrangling: filtering, missing values, binary Class label, one-hot encoding
- EDA: Matplotlib/Seaborn charts and SQL queries in SQLite
- Interactive visual analytics: Folium map and Plotly Dash dashboard
- Predictive analysis: StandardScaler → train/test split → GridSearchCV (LR, SVM, Decision Tree, KNN) → evaluate

Data Collection – SpaceX API



- Request: requests.get("https://api.spacexdata.com/v4/launches/past") → response.json()
- Helper functions turn IDs into BoosterVersion, LaunchSite + coordinates, PayloadMass + Orbit, and core details (Outcome, Flights, GridFins, Reused, Legs, LandingPad, Block, Serial)
- Missing PayloadMass replaced with the mean; LandingPad keeps None (26 rows) to mean "no landing pad used"

Data Collection – Web Scraping



- Page: "List of Falcon 9 and Falcon Heavy launches"; soup.title = List of Falcon 9 and Falcon Heavy launches - Wikipedia
- Third table = launch records; header cells cleaned with extract_column_from_header()
- Columns: Flight No., Date, Time, Version Booster, Launch site, Payload, Payload mass, Orbit, Customer, Launch outcome, Booster landing

Data Wrangling

- Launches per site: CCAFS SLC 40 = 55 · KSC LC 39A = 22 · VAFB SLC 4E = 13
- Orbits: GTO 27 · ISS 21 · VLEO 14 · PO 9 · LEO 7 · SSO 5 · MEO 3 · ES-L1, HEO, SO, GEO 1 each
- Class label: 1 = landed (True ASDS / True RTLS / True Ocean), 0 = otherwise → 60 vs 30 → 66.7% success
- One-hot encoding of Orbit, LaunchSite, LandingPad, Serial → 83 features, cast to float64

Outcome	Count	Class
True ASDS	41	1
None None	19	0
True RTLS	14	1
False ASDS	6	0
True Ocean	5	1
False Ocean	2	0
None ASDS	2	0
False RTLS	1	0

EDA with Data Visualization

- Scatter plots, coloured by landing outcome (Class), to see how two variables together relate to success:
  - Flight Number vs Launch Site · Payload Mass vs Launch Site
  - Flight Number vs Orbit · Payload Mass vs Orbit · Flight Number vs Payload
- Bar chart: success rate by orbit, to compare categories
- Line chart: yearly success rate, to show the trend over time
- Outcome: these patterns informed which features to keep for modelling

EDA with SQL

- Loaded Spacex.csv into SQLite (table SPACEXTBL, 101 rows) and queried with %sql
- Unique launch site names · 5 records where the site begins with "CCA"
- Total payload for NASA (CRS) · Average payload for booster F9 v1.1
- Date of the first successful ground-pad landing
- Boosters with a drone-ship success and payload 4,000–6,000 kg
- Count of successful vs failed mission outcomes
- Boosters that carried the maximum payload (subquery)
- 2015 drone-ship landing failures (month, booster, site)
- Ranking of landing outcomes between 2010-06-04 and 2017-03-20

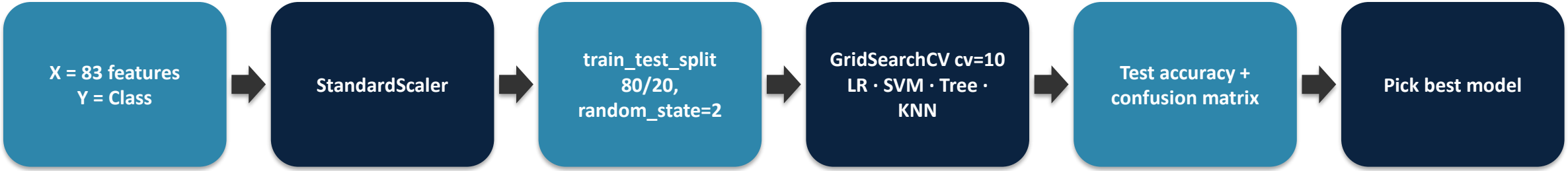
Interactive Map with Folium

- folium.Circle + label marker for each of the 4 launch sites (coordinates from the dataset)
- MarkerCluster of every launch: green = landed (Class 1), red = failed (Class 0)
- MousePosition to read coordinates of nearby features
- Haversine distance + PolyLine from a launch site to the nearest coastline, railway, highway and city
- Purpose: are sites near the equator and the coast? Are they close to transport links but far from cities?

Dashboard with Plotly Dash

- Dropdown: All Sites or a single launch site
- Pie chart: share of successes by site (All Sites), or success vs failure for the selected site
- RangeSlider: payload mass 0–10,000 kg
- Scatter plot: payload vs outcome, coloured by booster version category
- Two callbacks link the inputs to the charts
- Purpose: explore how site, payload and booster version affect success interactively

Predictive Analysis (Classification)

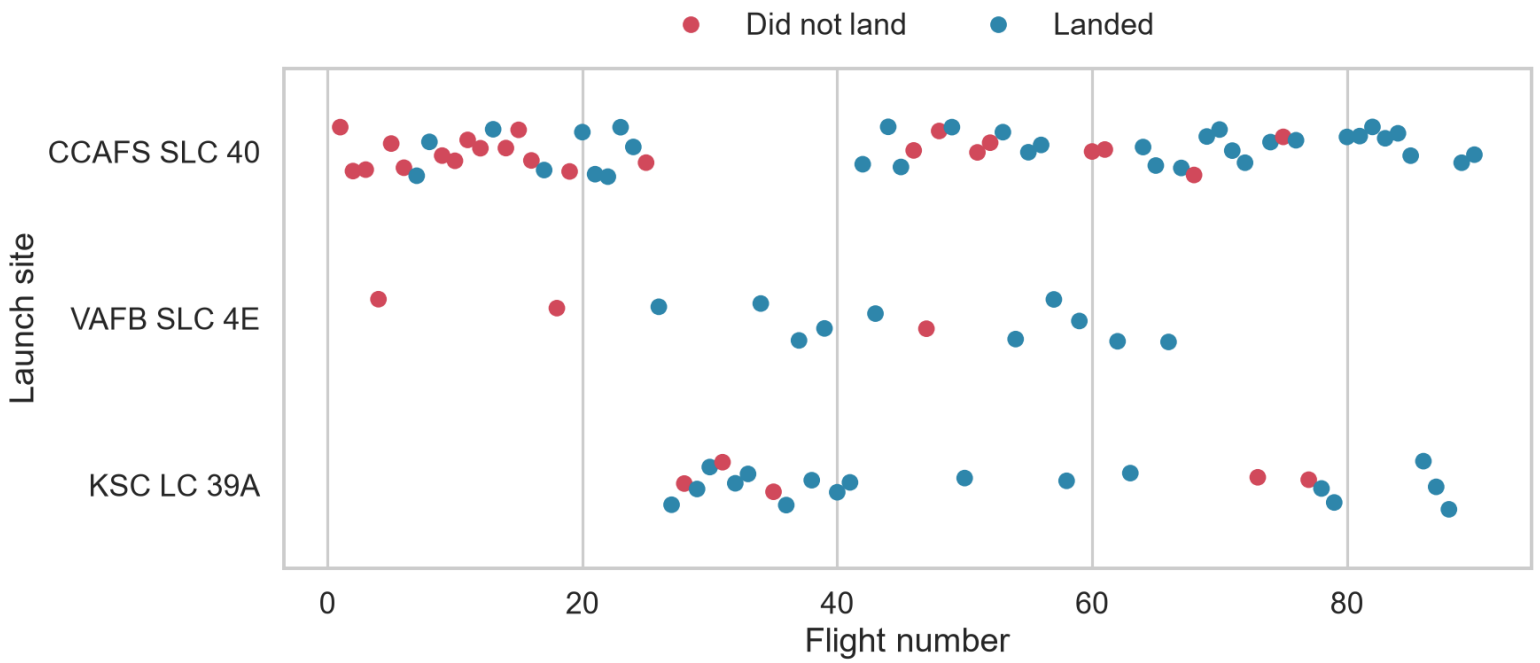


- 72 training and 18 test launches
- Logistic Regression: C, penalty, solver · SVM: kernel, C, gamma
- Decision Tree: criterion, splitter, max_depth, max_features, min_samples_leaf/split · KNN: n_neighbors, algorithm, p
- Compare best CV score, test accuracy and confusion matrix

Results

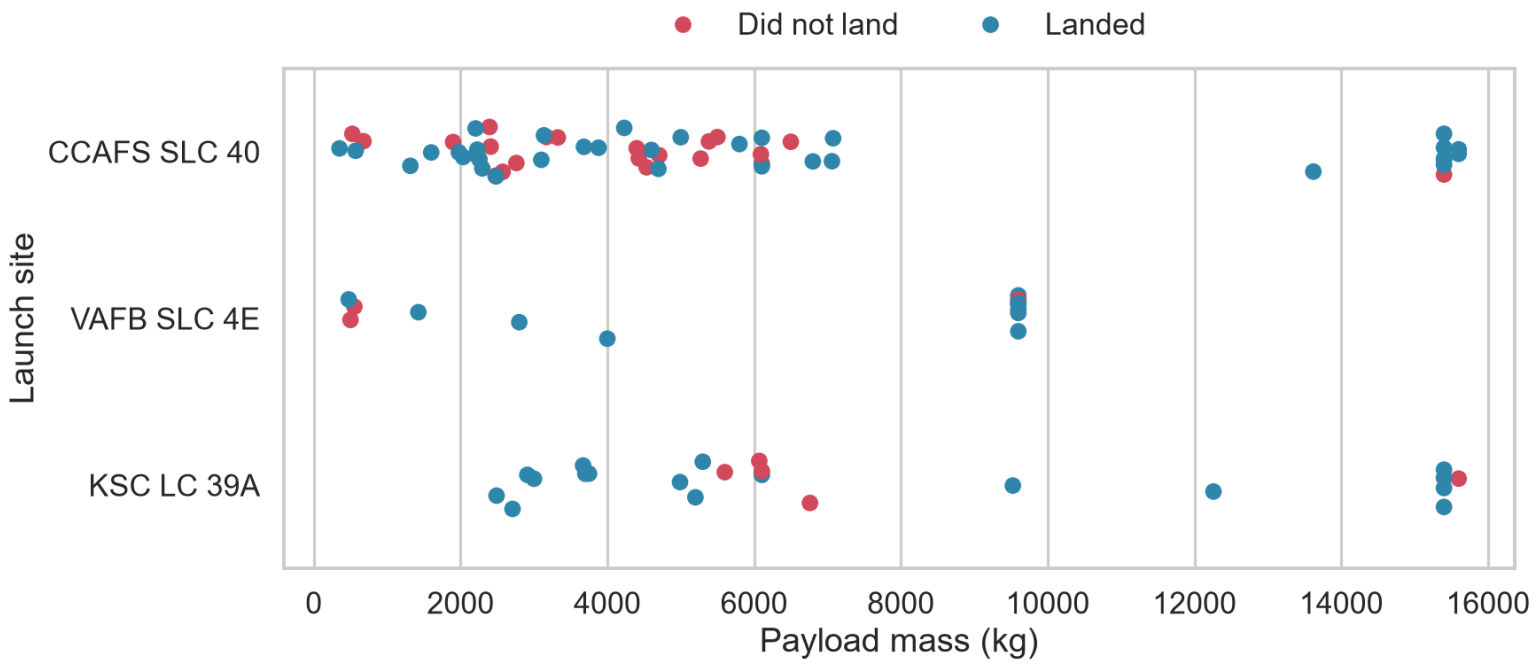
Exploratory data analysis · SQL · interactive analytics · predictive analysis

Flight Number vs Launch Site



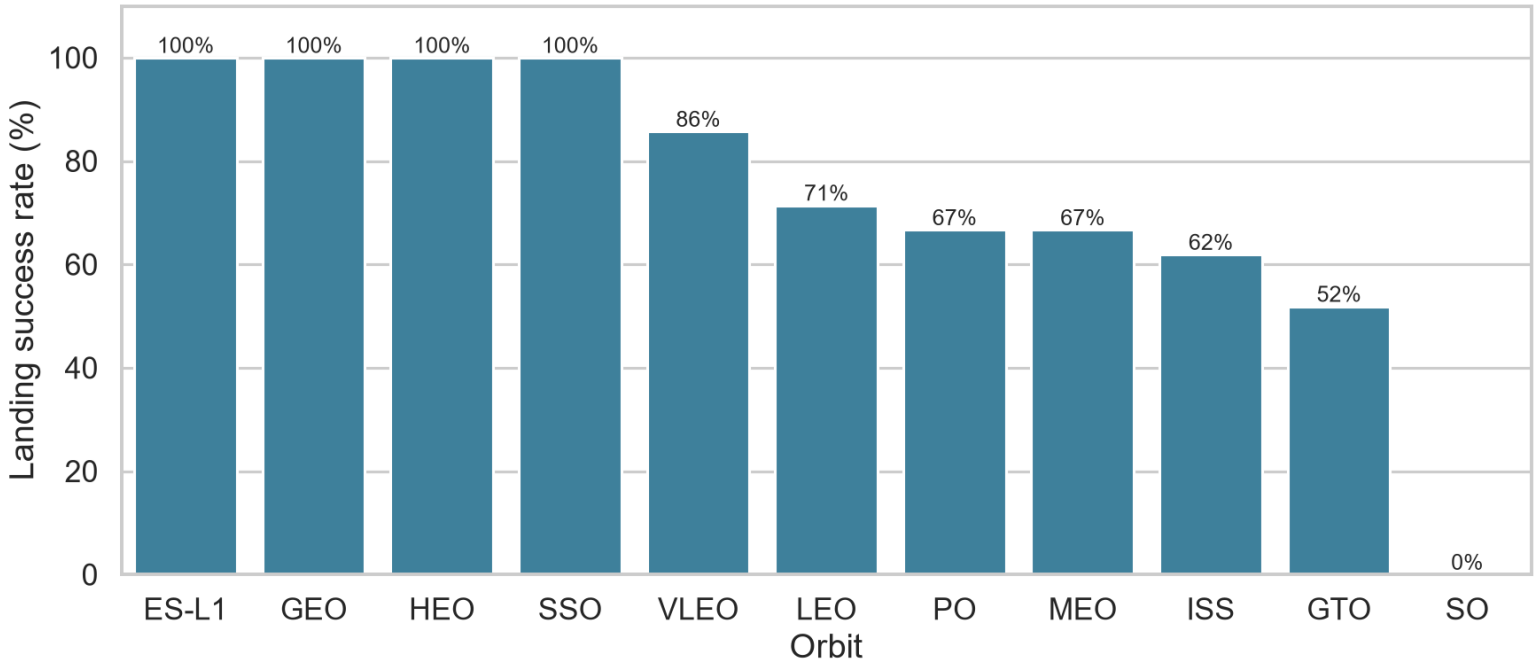
- Most early flights launched from CCAFS SLC 40, which also has the most failures
- VAFB SLC 4E was used from flight 4 to 66; KSC LC 39A from flight 27
- Success rises with flight number at every site
- Experience is a strong predictor

Payload vs Launch Site



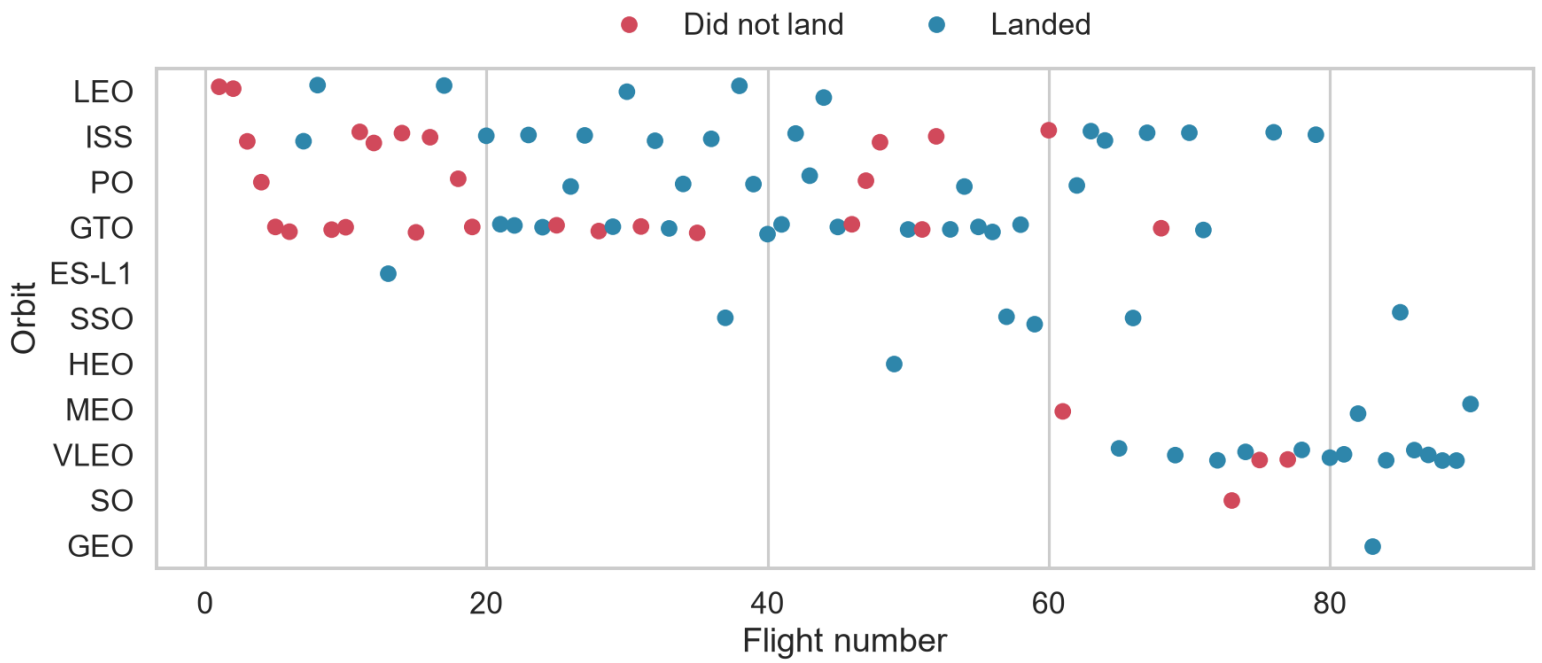
- CCAFS SLC 40: 350–15,600 kg
- KSC LC 39A: 2,490–15,600 kg
- VAFB SLC 4E: no launches above 10,000 kg (max 9,600 kg)
- Heavy payloads (>10,000 kg) mostly landed

Success Rate by Orbit Type



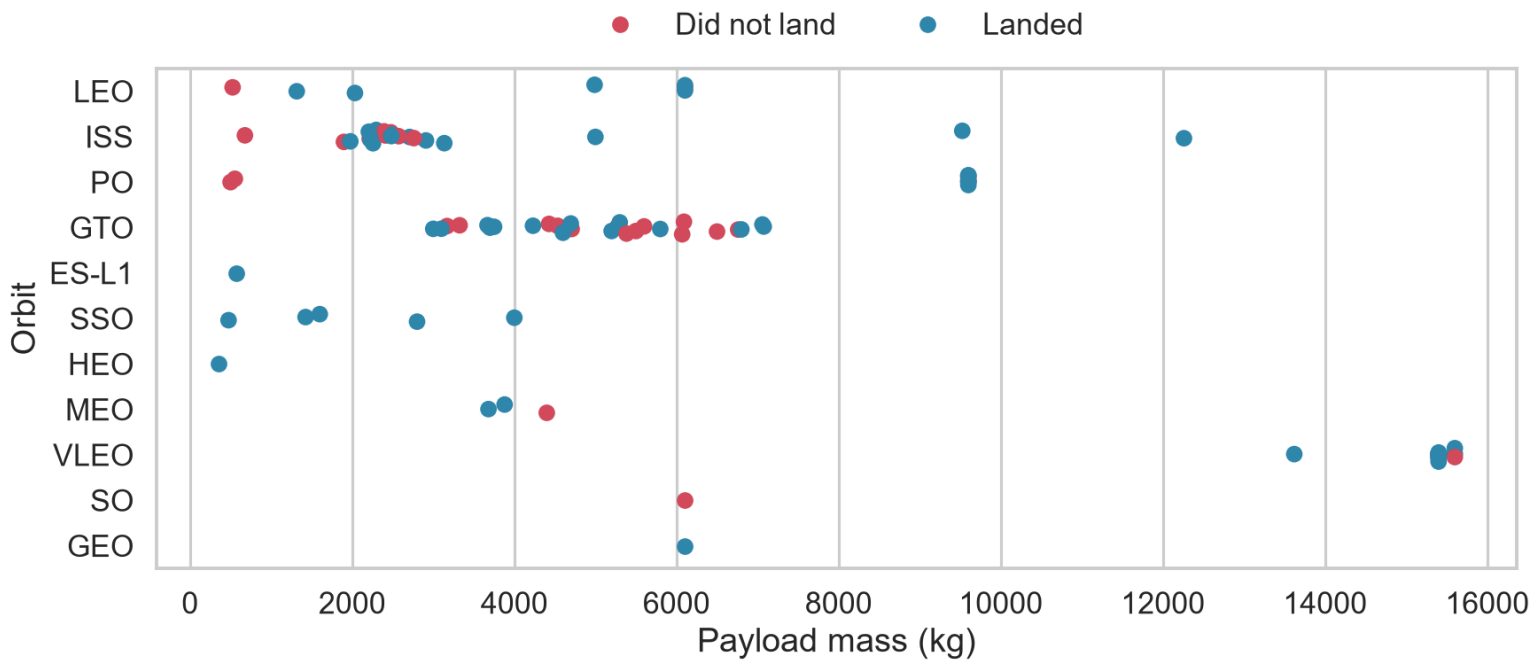
- 100%: ES-L1, GEO, HEO (1 launch each), SSO (5/5)
- VLEO 85.7% · LEO 71.4% · PO 66.7% · MEO 66.7% · ISS 61.9%
- GTO lowest of the common orbits: 51.9% of 27
- SO 0% (1 launch)
- Orbits with 1 launch say little

Flight Number vs Orbit Type



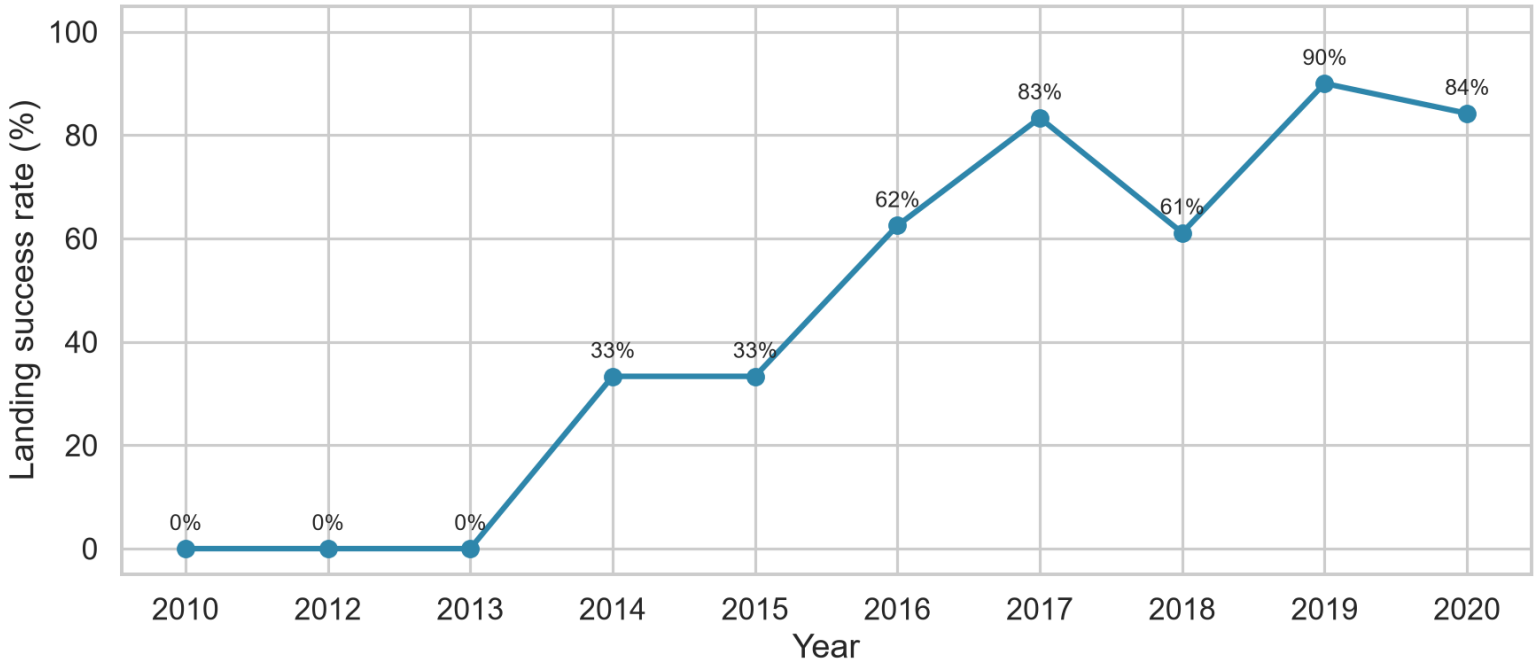
- Early flights: LEO, ISS and GTO
- VLEO (Starlink) appears from flight 65 and mostly lands
- LEO: success appears related to flight number
- GTO: no clear relationship with flight number

Payload vs Orbit Type



- Payloads above 10,000 kg: almost all VLEO (plus one ISS at 12,259 kg)
- Heavy payloads land more often for PO, LEO and ISS
- GTO (3,000–7,076 kg): mixed, no clear pattern
- SSO ≤ 4,000 kg, all landed

Launch Success Yearly Trend



- 2010–2013: 0%
- 2014–2015: 33%
- 2016: 62% · 2017: 83%
- 2018: 61% (a dip)
- 2019: 90% · 2020: 84%
- Clear upward trend since 2013

All Launch Site Names


```
SELECT DISTINCT Launch_Site FROM SPACEXTBL;
```

Launch_Site	Launches in table
CCAFS LC-40	26
CCAFS SLC-40	34
KSC LC-39A	25
VAFB SLC-4E	16

Launch Sites Beginning with 'CCA'


```
SELECT * FROM SPACEXTBL WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

Date	Booster	Site	Payload	kg	Orbit	Customer	Landing
2010-06-04	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Un	0	LEO	SpaceX	Failure (parachute)
2010-12-08	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSat	0	LEO (ISS)	NASA (COTS) NRO	Failure (parachute)
2012-05-22	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	No attempt
2012-10-08	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	No attempt
2013-03-01	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	No attempt

Total Payload Mass – NASA (CRS)


```
SELECT SUM(PAYLOAD_MASS__KG_) AS total_payload
FROM SPACEXTBL WHERE Customer = 'NASA (CRS)';
```

- Result: 45,596 kg
- Total payload carried for NASA's Commercial Resupply Services missions

Average Payload Mass – F9 v1.1


```
SELECT AVG(PAYLOAD_MASS__KG_) AS avg_payload
FROM SPACEXTBL WHERE Booster_Version = 'F9 v1.1';
```

- Result: 2,928.4 kg
- Average payload for the F9 v1.1 booster version

First Successful Ground-Pad Landing


```
SELECT MIN(Date) AS first_success
FROM SPACEXTBL WHERE Landing_Outcome = 'Success (ground pad)';
```

- Result: 2015-12-22
- Orbcomm OG2 mission from CCAFS LC-40

Drone-Ship Success with Payload 4,000–6,000 kg


```
SELECT Booster_Version FROM SPACEXTBL
WHERE Landing_Outcome = 'Success (drone ship)'
  AND PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_ < 6000;
```

- F9 FT B1022 · F9 FT B1026 · F9 FT B1021.2 · F9 FT B1031.2
- All four are Full Thrust boosters; two are re-flown cores (.2)

Total Successful and Failed Mission Outcomes


```
SELECT Mission_Outcome, COUNT(*) AS total
FROM SPACEXTBL GROUP BY Mission_Outcome;
```

- Success: 99 · Success (payload status unclear): 1 · Failure (in flight): 1
- → 100 successful missions, 1 failure
- Mission success is near-perfect; landing the booster is the harder problem

Boosters Carrying the Maximum Payload


```
SELECT Booster_Version FROM SPACEXTBL WHERE PAYLOAD_MASS__KG_ =
  (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTBL);
```

- Maximum payload: 15,600 kg
- F9 B5 B1048.4 · B1049.4 · B1051.3 · B1056.4 · B1048.5 · B1051.4
- F9 B5 B1049.5 · B1060.2 · B1058.3 · B1051.6 · B1060.3 · B1049.7
- Only Block 5 boosters, mostly re-flown, carried the maximum payload

2015 Drone-Ship Landing Failures


```
SELECT substr(Date,6,2) AS month, Landing_Outcome, Booster_Version, Launch_Site
FROM SPACEXTBL WHERE Landing_Outcome = 'Failure (drone ship)'
  AND substr(Date,1,4) = '2015';
```

Month	Landing outcome	Booster	Launch site
01 (January)	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04 (April)	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Landing Outcomes Ranked (2010-06-04 to 2017-03-20)

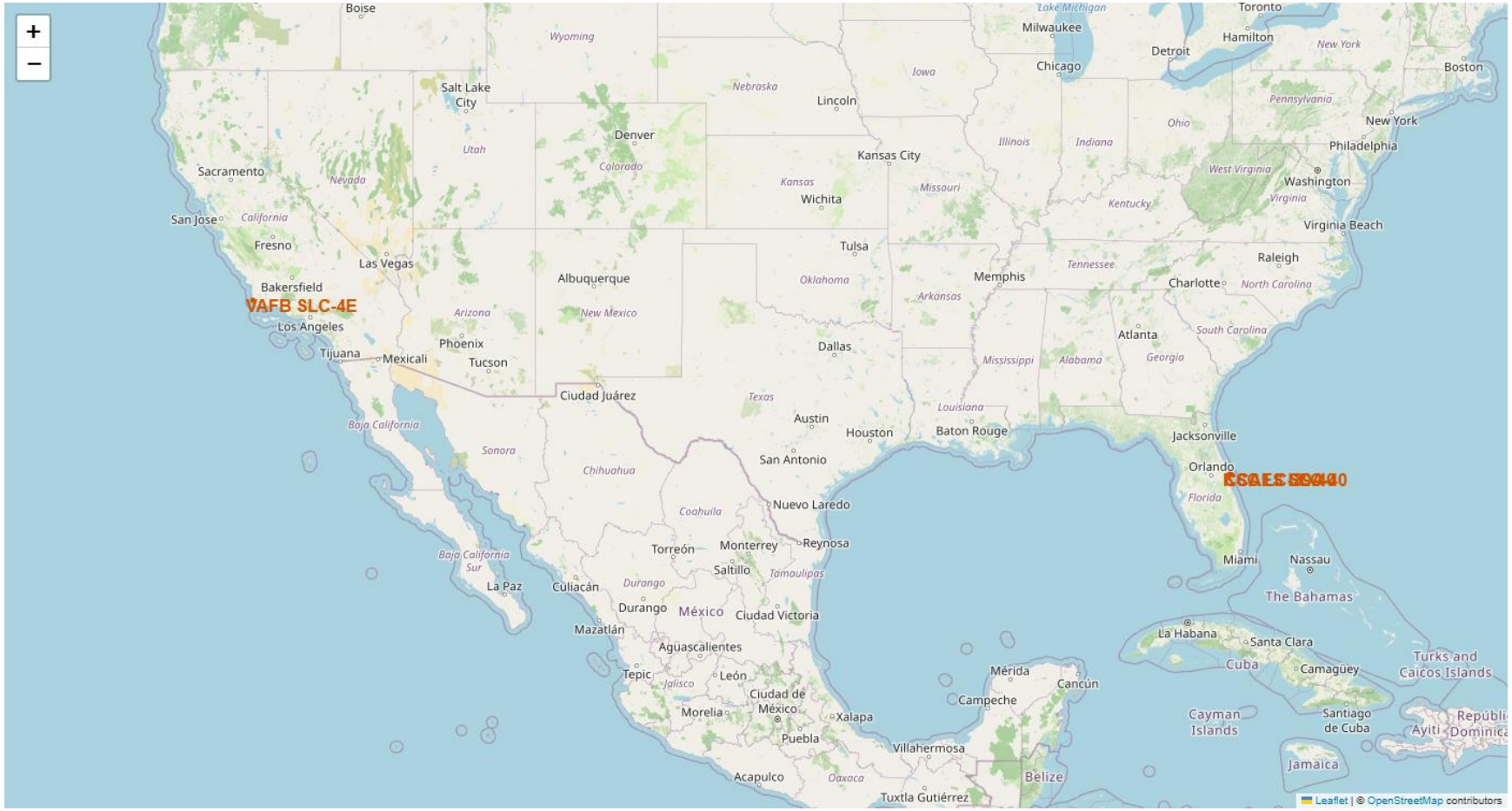

```
SELECT Landing_Outcome, COUNT(*) AS n FROM SPACEXTBL
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome ORDER BY n DESC;
```

Rank	Landing outcome	Count
1	No attempt	10
2	Success (drone ship)	5
3	Failure (drone ship)	5
4	Success (ground pad)	3
5	Controlled (ocean)	3
6	Uncontrolled (ocean)	2
7	Failure (parachute)	2
8	Precluded (drone ship)	1

Launch Site Proximity Analysis

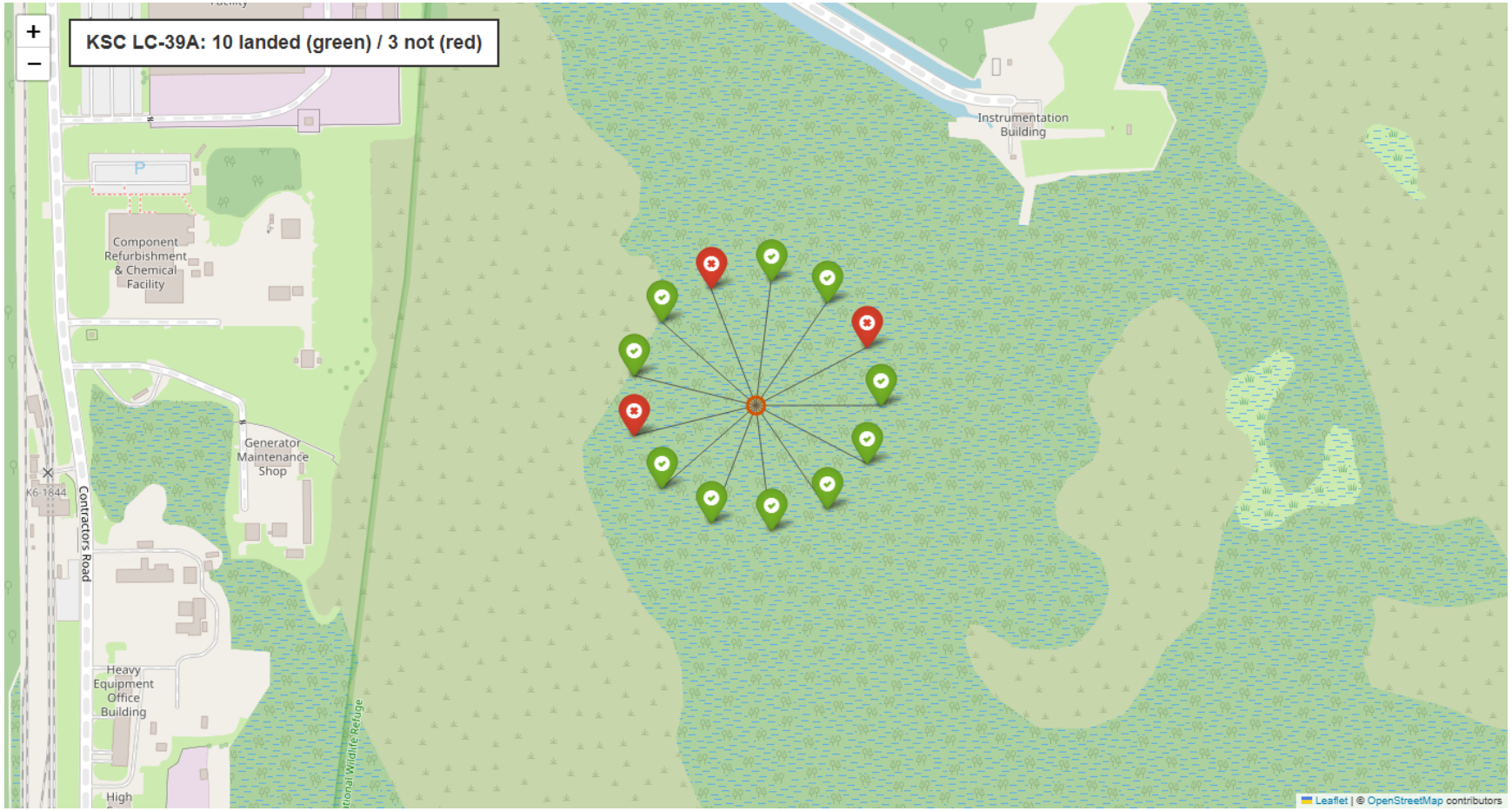
Folium interactive map

All Launch Sites on a Map



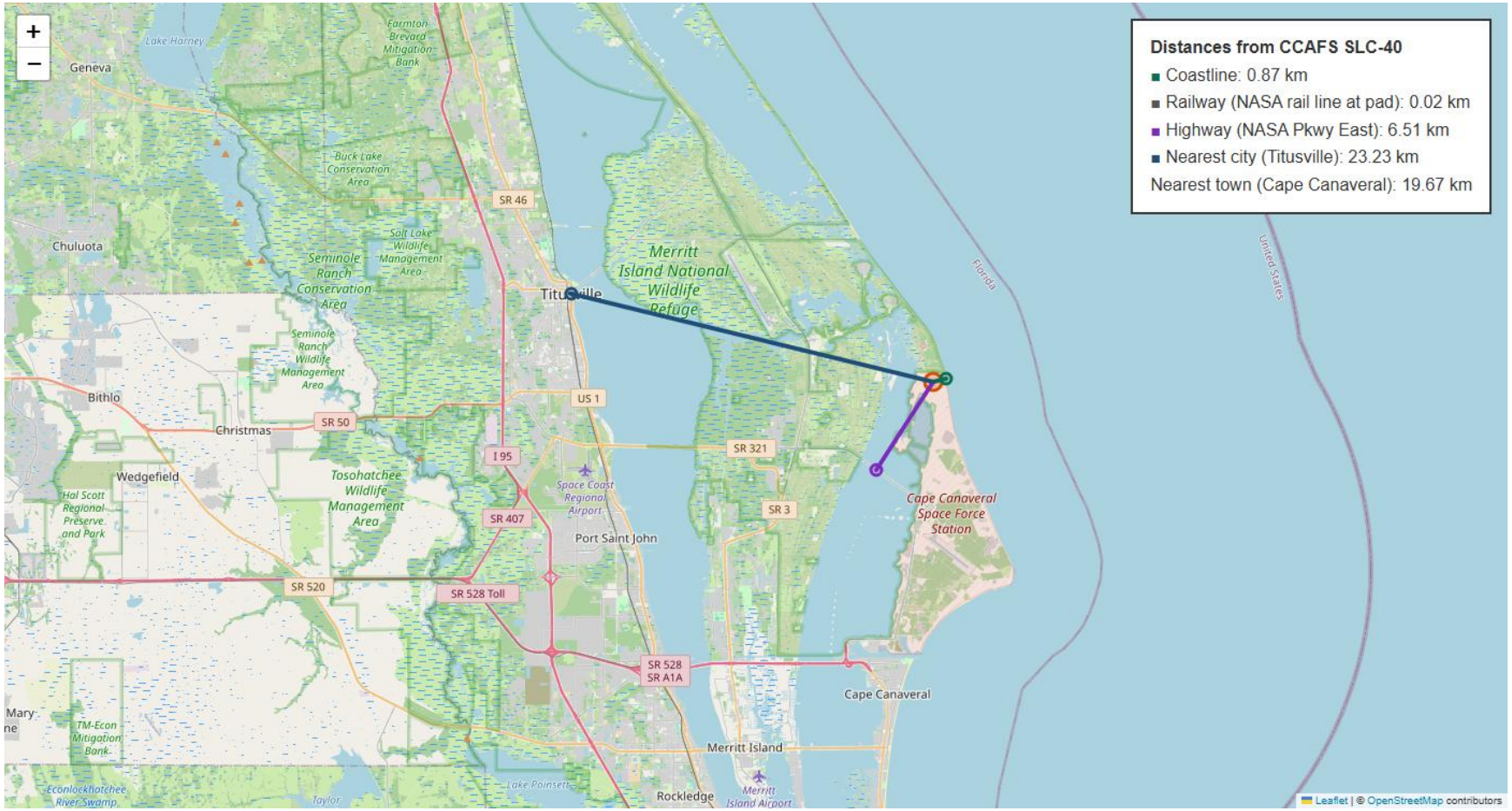
- 3 sites in Florida (CCAFS LC-40, CCAFS SLC-40, KSC LC-39A), 1 in California (VAFB SLC-4E)
- All sites are on the coast, so rockets fly over the ocean
- All are in the far south of the US (Florida ≈ 28.5° N), as close to the equator as the mainland allows

Colour-Labelled Launch Outcomes



- Each launch is a marker: green = landed, red = failed
- KSC LC-39A (shown): 10 of 13 landed, the best record of any site
- CCAFS LC-40: 7 of 26 landed (many early flights)
- VAFB SLC-4E: 4 of 10 · CCAFS SLC-40: 3 of 7

Distances to Coastline, Railway, Highway and City

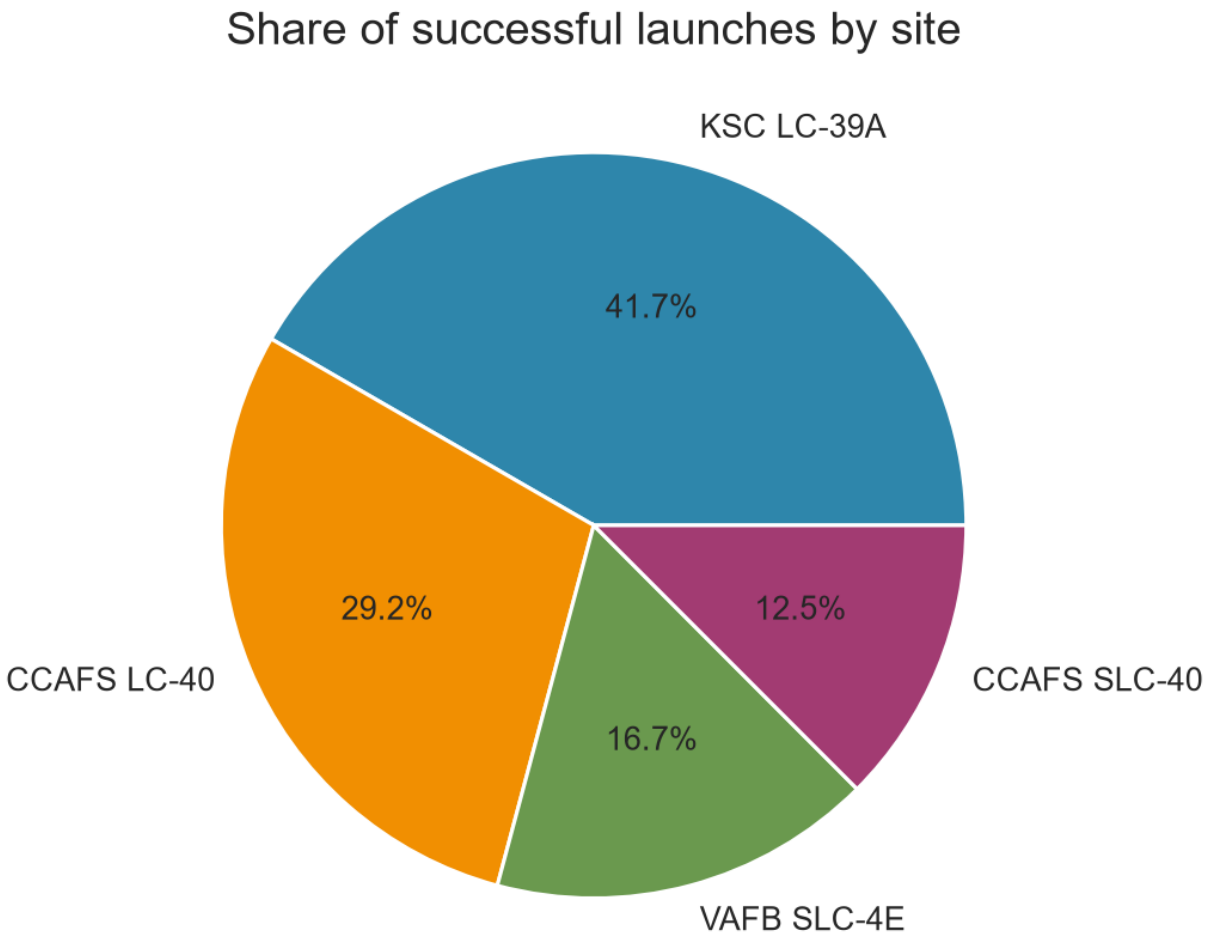


- From CCAFS SLC-40:
  - Coastline: 0.87 km
  - Railway: 0.02 km (rail line runs to the pad)
  - Highway (NASA Pkwy East): 6.51 km
  - Nearest city (Titusville): 23.23 km
- Close to coast, rail and road (logistics); far from cities (safety)

Build a Dashboard with Plotly Dash

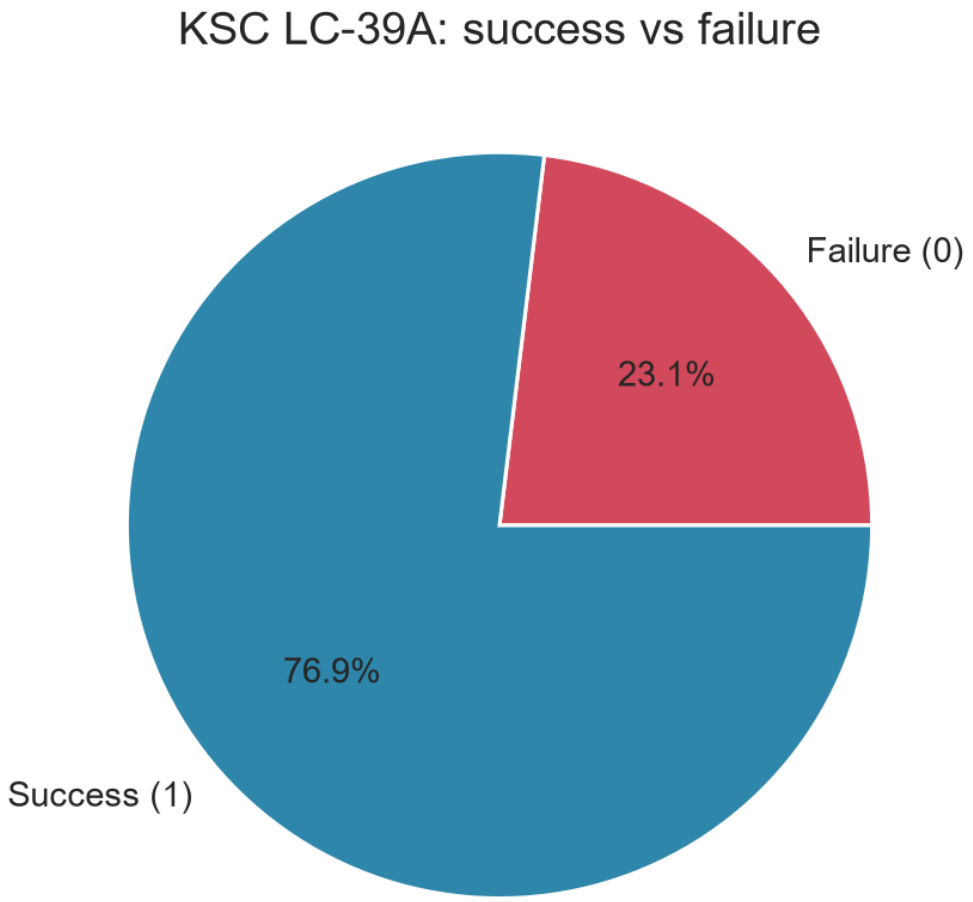
Interactive dashboard results

Launch Success Count for All Sites



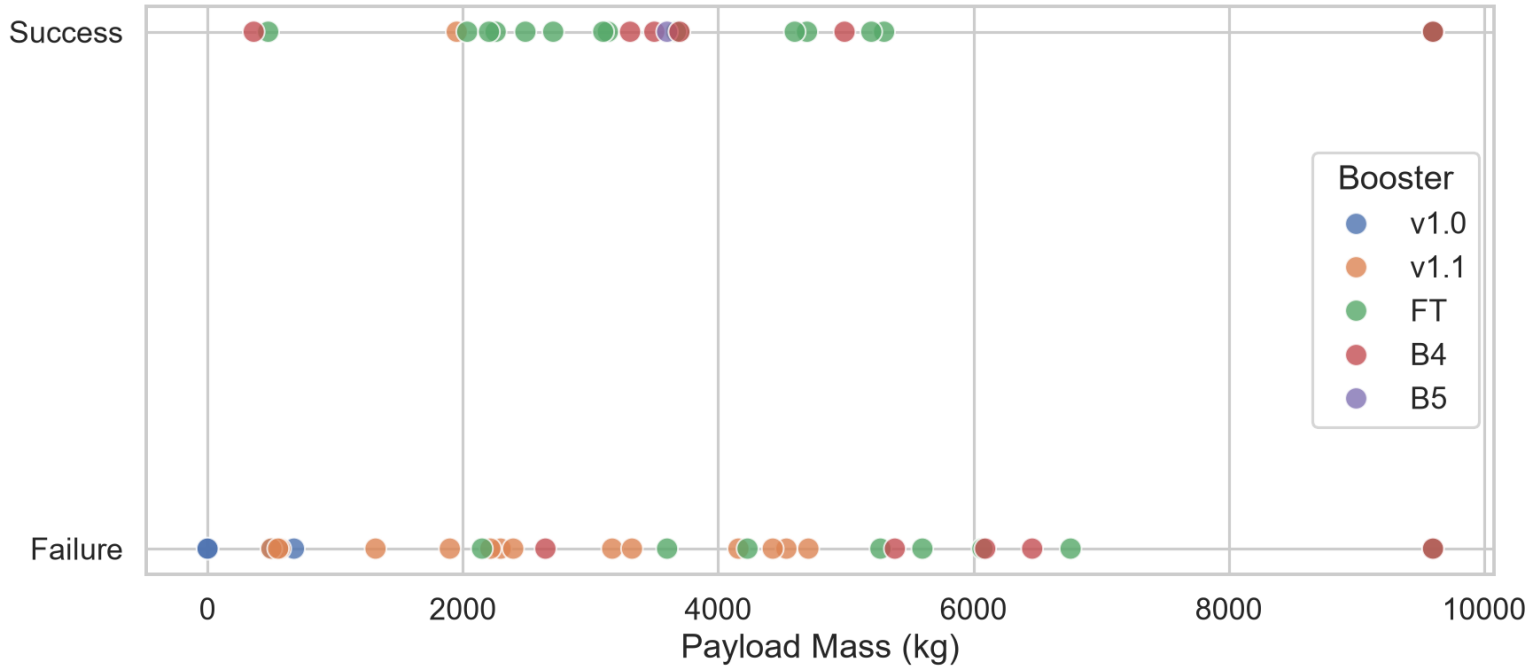
- KSC LC-39A: 41.7% of all successful launches
- CCAFS LC-40: 29.2%
- VAFB SLC-4E: 16.7%
- CCAFS SLC-40: 12.5%
- KSC LC-39A leads the success count

Launch Site with the Highest Success Ratio



- KSC LC-39A: 76.9% success (10 of 13)
- 23.1% failure (3 of 13)
- The highest success ratio of any site

Payload vs Launch Outcome (All Sites)

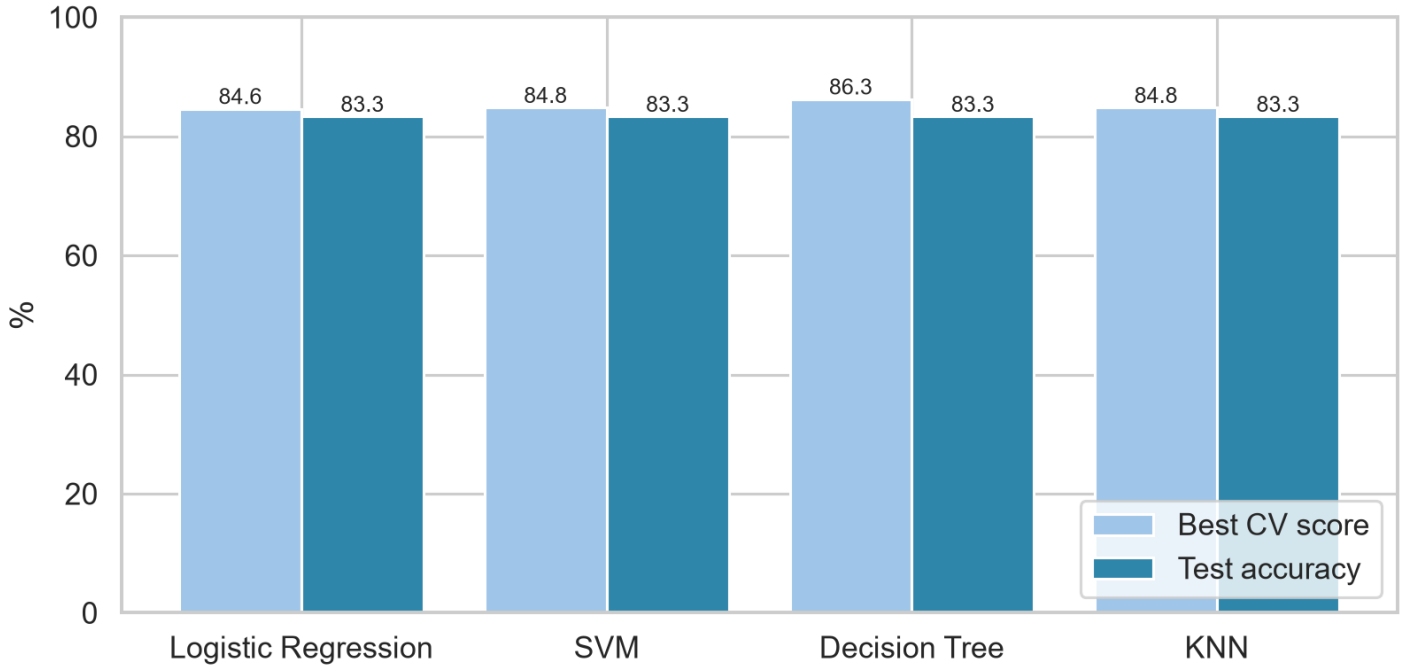


- Success by booster: v1.0 0% (0/5) · v1.1 7% (1/15) · FT 67% (16/24) · B4 55% (6/11) · B5 100% (1/1)
- Success by payload: 0–2k kg 23% · 2–4k kg 62% · 4–6k kg 38% · 6–10k kg 33%
- FT has the best rate among well-used boosters; 2,000–4,000 kg is the best payload range

Predictive Analysis (Classification)

Model results

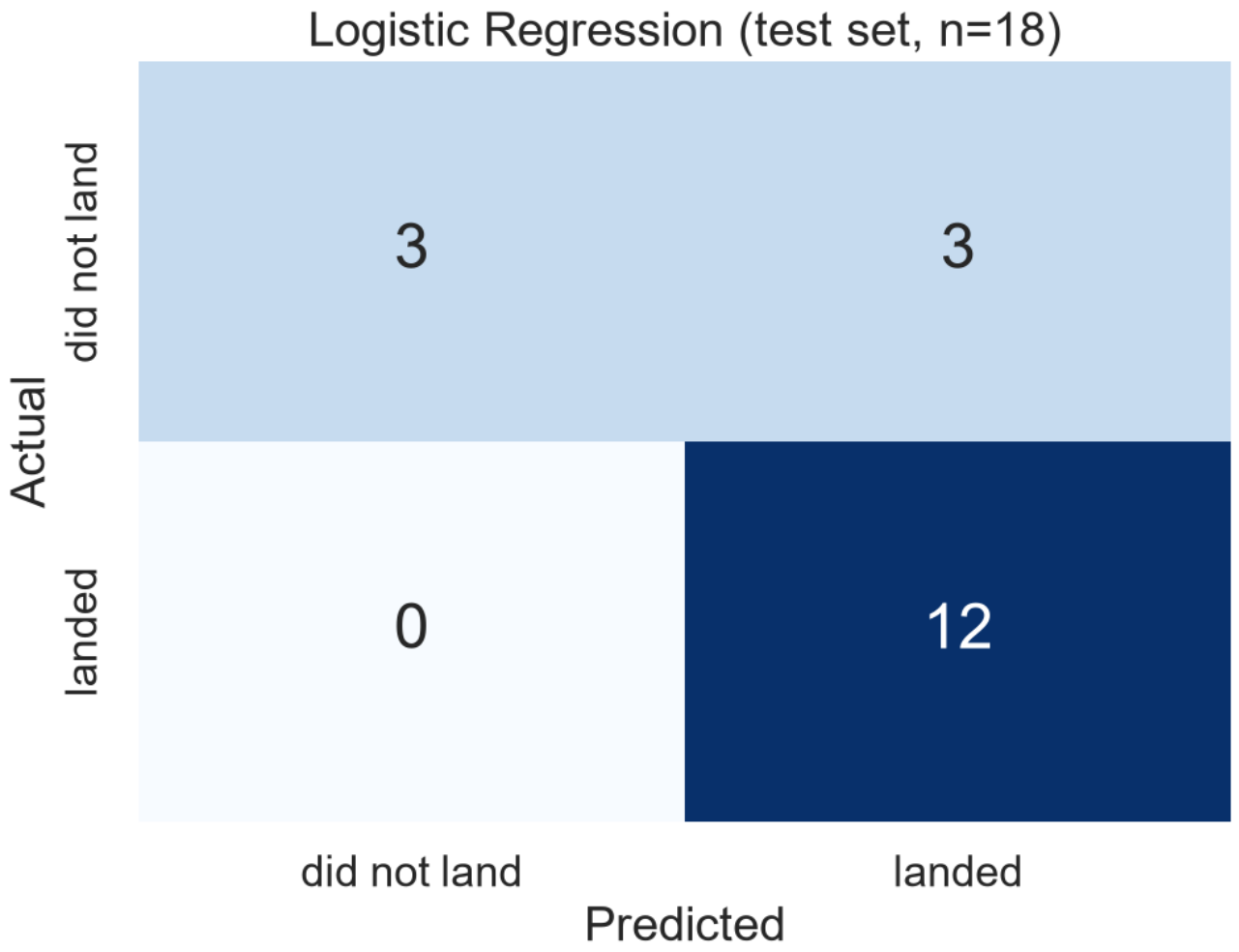
Classification Accuracy



Model	Best CV	Test
Logistic Regression	84.6%	83.3%
SVM	84.8%	83.3%
Decision Tree	86.3%	83.3%
KNN	84.8%	83.3%

- All four tie on test accuracy (15/18)
- Best model: Decision Tree, with the highest CV score (86.3%)
- With 18 test launches, one prediction = 5.6 points, so the models are effectively equal

Confusion Matrix



- All four models give the same matrix on the test set
- 12 landings correctly predicted, 0 missed (recall 100%)
- 3 failures correctly predicted
- 3 failures wrongly predicted as landings (false positives)
- The model is optimistic: false positives are the main weakness

Conclusions

- Landing success follows a learning curve: 0% (2010–13) → 84–90% (2019–20)
- KSC LC-39A has the highest success rate (76.9%); newer boosters (FT, B5) land far more often
- Orbit matters: SSO and VLEO are reliable, GTO is about 50/50
- Tuned classifiers reach 83.3% test accuracy; Decision Tree is selected (best CV score, 86.3%)
- Insight for a competitor: recent, non-GTO missions on newer boosters → expect a reused (cheaper) first stage. Treat landing predictions for GTO cautiously because of false positives.
- Next steps: post-2020 launches, a larger test set, and features such as booster reuse count and weather

Appendix

- GitHub repository: https://github.com/Downey112/ibm-ds-capstone-spacex
  - Notebooks: SpaceX API, web scraping, data wrangling, EDA visualisation, EDA SQL, Folium map, Plotly Dash app (.py), ML prediction
- Datasets: modelling dataset 90 rows · SQL table 101 rows · Dash dataset 56 rows
- Best hyperparameters:
  - Logistic Regression: C=0.01, penalty=l2, solver=lbfgs
  - SVM: C=1.0, gamma=0.0316, kernel=sigmoid
  - Decision Tree: criterion=gini, max_depth=6, max_features=sqrt, min_samples_leaf=4, min_samples_split=10, splitter=random
  - KNN: algorithm=auto, n_neighbors=10, p=1
- Proximity distances: OpenStreetMap features + haversine formula